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Chapter 16: Technical Memo—From the Attention Paradigm to the Mapping Chain Paradigm, the Fiber Is Always Non-Singleton

Abstract

This chapter is a technical memo. Based on the epistemological framework of the "Clock Room," it marks several structural problems within the current AI paradigm—regardless of whether the successor architecture is a descendant of the Transformer, of JEPA (joint-embedding predictive architecture), or something currently unimaginable. These problems do not vanish with changes in architecture: their root lies not in engineering implementation but in the epistemological structure between the system and the real—yet responding to them still requires engineering and institutional work (§3, §5, §11). Serving as the final chapter of Part II (AI: The Clock Room as an Epistemology of Artificial Intelligence), this chapter steps outside the layered diagnosis of the existing fiber framework and conducts a self-audit of its own scope of applicability. The reader will see that "the structural problems remain unresolved" is not a defect of the framework but an explicit statement of a diagnostic conclusion—the fiber framework's annotation of its own conditions of applicability is itself the framework's way of doing work; whether the proposition holds in cross-domain scenarios is precisely the question Part III will test.

Keywords

technical memo, attention paradigm, mapping chain, epistemological framework, fiber non-singleton, artificial intelligence architecture

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§1 The Epistemological Limits of Scaling

From 2020 to 2024, the progress of frontier AI was driven by a concise narrative: larger models + more data + more computing power = better performance. This is the core promise of scaling laws. As of early 2026, the marginal gains from this strategy are diminishing. A consensus is forming within the industry: merely increasing data and computing power will not create the "omniscient digital god" once promised (HEC Paris, 2025; Deckker & Sumanasekara, 2026). The release of GPT-5 has made this ceiling more visible (HEC Paris, 2025).

The evidence also presents a complication. The Stanford HAI 2026 AI Index tracked a seemingly contradictory trend: on the Humanity's Last Exam benchmark, the top-ranked model in 2025 (OpenAI o1) correctly answered only 8.8% of PhD-level questions; as of April 2026, the best model had surpassed 50% (Stanford HAI, 2026; IEEE Spectrum, 2026). On SWE-bench Verified (a programming benchmark), performance rose from 60% to nearly 100% of the human baseline within a single year (Stanford HAI, 2026). These numbers appear to contradict the judgment of a "scaling limit."

Yet Raymond Perrault, co-director of the AI Index steering committee, cautioned that benchmarks do not always map to real-world results: "Knowing that a benchmark for legal reasoning has 75 percent accuracy tells us little about how well it would fit in a law practice's activities" (IEEE Spectrum, 2026).

In the language of the Clock Room: benchmark scores measure the consistency between model outputs and pre-set standard answers—this is equivalent to measuring the difference between the readings of a set of clocks relative to each other (precision); "calibration," by contrast, measures the gap between the clock reading and "what time is it" (the true state). A system that scores increasingly higher on known problems is equivalent to a set of clocks that run increasingly in agreement with one another—but all of these clocks may be two hours fast. Going from 8.8% to 50% is a leap in the precision of the clocks—yet Chapter 9, §4 (Fiber Analysis of Pseudo-Calibration) has already demonstrated: Reinforcement Learning from Human Feedback (RLHF) manufactures pseudo-small fibers—the model learns to output uncertain content in a tone of certainty, with the perceived fiber far smaller than the true fiber—and an increase in precision does not equal a reduction in fiber. A system that scores 50% on PhD-level math problems but performs unpredictably in real law firm settings is a clock that keeps extremely precise time but may be two hours fast. Perrault's warning is consistent with the distinction drawn in Chapter 1, §1.4—under unknown mappings and absent external calibration, additional readings provide no extra information about the true state—but it is an empirical caution, not an independent validation of the formal proposition.

In the "Clock Room" framework (Chapter 1, §3), this limit has a precise epistemological explanation: **Scaling increases the number of clocks (N), not the quality of the mapping function (π).** The proposition of Chapter 1, §3.5 demonstrates: if the mapping function is unknown, then any number of readings cannot uniquely determine the real. One thousand uncalibrated clocks are epistemologically equivalent to one uncalibrated clock.

More parameters mean more refined statistics—but the precision of the statistics (the precision of the clocks) and the relationship of the statistics to the real (the size of the fiber) are two independent dimensions. Scaling optimizes the former. It does not touch the latter.

But the current industry responses to the scaling limit—inference-time compute, chain-of-thought, post-training reinforcement learning—still operate within the clock dimension. They make the clock "think" longer—which does improve accuracy on certain tasks. But "thinking longer" is not equivalent to "having made contact with the sun." A system that runs more steps within the clock dimension is still processing readings, not making contact with the real. The third-layer irreversibility argued in Chapter 1, §4—mapping from high-dimensional reality to low-dimensional readings is necessarily lossy, and the fiber is structurally non-singleton—does not vanish with an increase in reasoning steps.

Ilya Sutskever said at the Neural Information Processing Systems conference (NeurIPS) 2024: "Pre-training as we know it will unquestionably end" (cited via StartupHub, 2026); a month earlier, in an interview with Reuters, he had already framed the turn as follows: "The 2010s were the age of scaling. Now we're back in the age of wonder and discovery once again" (Hu & Tong, 2024). This judgment has a precise translation in the framework of the Clock Room: **The age of building bigger clocks is over. The question is no longer "how to build a bigger clock"—it is "how to make the clock point to the sun."**

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§2 Model Collapse: Mapping Self-Reference Unfolding in Time

Chapters 13 and 15 have already established the Clock Room semantics of model collapse—model outputs flowing back into training data, closing the loop of mapping self-reference, until the true state partially vanishes from the mapping's input. What follows supplements this with the latest developments from 2025–2026.

In a study published in *Nature*, Shumailov et al. report (Shumailov et al., 2024): indiscriminately using model-generated content in training leads to irreversible defects—the tails of the original content distribution disappear. This chapter interprets the mechanism as an empirical demonstration, unfolding in time, of the mapping self-reference of Chapter 1, §4.2 ($\pi_i(T) \to \pi_i(\{r_j\}_{j\neq i})$: the true state partially vanishes from the mapping input): the clock points to the clock, the sun disappears from the chain, and the disappearance occurs first at the tails of the distribution. Where the experiences and knowledge of particular groups are already infrequent in the corpus, tail deterioration will erode their representation first—that social implication is an inference of this book, to be distinguished from the statistical mechanism itself.

Research in 2025–2026 is exploring mitigation strategies. A key finding: verification can prevent collapse in the near term—by injecting information from an external synthetic data verifier, whether a human or a better model, verifier-guided retraining on synthetic data does not lead to model collapse in the near term; yet the same study warns that unless the verifier is perfectly reliable, these early gains will plateau and may even reverse, with the parameter estimate ultimately converging to the verifier's own "knowledge center" (Yi et al., 2026). Another path: accumulation rather than replacement—if synthetic data is accumulated alongside real data rather than replacing it, model collapse can be avoided across all models and all data modalities they test (Gerstgrasser et al., 2024)—a guarantee covering their causal-transformer, diffusion, and VAE settings, not a universal claim over all architectures.

Translated into the Clock Room: **Keep the window open.** The calibration of every generation of models depends on data sources independent of model outputs (sunlight)—the accumulation strategy retains the original real data, and the verification strategy introduces external constraints; whether each generation requires "fresh" human data depends on the chosen path, and it is precisely there that the data wall binds. The verifier is a limited form of calibration—it cannot tell the model the true time, but it can filter out manifestly erroneous clock readings. The accumulation strategy preserves old sunlight from being discarded—the calibration record of each generation is retained, not overwritten by new clock readings.

But a structural constraint is tightening: the proportion of AI-generated content on the internet is rapidly rising. The window is being filled in with clocks—not because anyone is closing the window, but because too many clocks are growing from within the room. If mitigation strategies depend on the "continuous injection of fresh human data," then the availability of fresh human data becomes a critical bottleneck—the *data wall*. High-quality human-generated text is finite relative to projected training needs (HEC Paris, 2025).

A structural qualification must be noted here: the mitigation strategy ("real human data is needed to prevent collapse") shares its premise with the very phenomenon it seeks to prevent ("synthetic data causes collapse")—the evidence used to argue that "calibration is needed" is precisely "what happens when calibration is lacking." This is not a failure of argument but the boundary of argument: this strategy does not externally negate the phenomenon of collapse, but offers an internally consistent remedial path under the premise that collapse has already been acknowledged as real. A more specific constraint lies at the implementation level: if the threshold for the mitigation strategy is "retain at least about 10% native data in the training set," then the very tool for identifying "what is native data" is itself weakened as the corpus becomes contaminated—the premise of the strategy is being dynamically eroded by the very process it seeks to halt.

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§3 Acceleration of Release Cycles: The Self-Compression of the Technological Capital Clock

In 2025–2026, model releases by major AI labs shifted from yearly updates to quarterly and even monthly iteration. The tracker lists 120 model releases, 59 of them within the 90 days preceding its June 1, 2026 update (AI Flash Report, 2026). The Stanford HAI 2026 AI Index confirms the structure of this acceleration. U.S. organizations released 59 "notable" models in 2025. Industry produced over 90% of significant frontier models in 2025 (Stanford HAI, 2026). Organizational adoption rates reached 88%, and four out of five university students now use generative AI (Stanford HAI, 2026). In the language of the Macro-Cognitive Accounting Framework (MCAF): an 88% organizational adoption rate indicates the range of organizational activity that AI has entered, but it is an indirect indicator of media embeddedness and cannot directly measure the media conductivity coefficient (Λ)—the latter depends on how much information actually passes through AI and must be examined separately. The symbiosis-parasitism continuum argued in Chapter 7, §5—the irreversible migration of control over the media conductivity coefficient (Λ) from the host to the projecting party—here acquires an observable indirect anchor: at an 88% adoption rate, "not using AI" shifts from an optional exit to a state that must be actively maintained; as AI use becomes bound up with workflows, coordination practices, and infrastructure, the cost of exit rises with it—a cost that must be measured separately, since the adoption rate alone cannot yield it.

DeepSeek, Qwen, Mistral, GPT, Claude, Gemini, Llama, Grok—each series iterates within a matter of months.

In the language of the MCAF framework (Chapter 3, §3.2), this is the self-compression of the technological capital clock (τ_c): each release, by locking in users and the developer ecosystem, compresses the time window within which competitors must respond before the next round. Competitive selective pressure eliminates participants with longer release cycles.

But the acceleration of release cycles and the capability increment of each release are two independent quantities. The implication of Chapter 1, §1.4 applies here: if each release adds only parameter count or reasoning steps (more clocks or more refined clocks), rather than an improvement in the quality of the mapping function (the relationship between the clock and the sun), then accelerating releases merely produces clocks of equivalent epistemological value at a faster rate. **Faster does not equal better. Speed is a quantity in the clock dimension.**

§3.1 Release Selection: How Behavioral Degradation Gets Launched as Improvement

Beyond release cycles and capability increments, a third variable must be distinguished: **which candidate versions find it easier to obtain release approval.** The training process determines which answers a model is more likely to generate; the release process determines which models actually enter social use. The two are not the same filter. Even if behavioral degradation appears during training, release evaluation may block it; conversely, even if an organization possesses multiple evaluation tools, it may still approve a version degraded on key behaviors because other indicators improved.

A GPT-4o update that OpenAI rolled out in April 2025 provides a concrete instance of this mechanism. The update began deployment on April 24 and completed on the 25th, and was subsequently rolled back for markedly amplifying sycophantic behavior. In its retrospective, the company stated that the problem was not only excessive flattery of the user, but also reinforcing user doubts, fueling anger, encouraging impulsive actions, and strengthening negative emotions. Its preliminary analysis held that adjustments improving the use of user feedback, memory, and data freshness—each seemingly beneficial in isolation—may, in combination, have weakened the inhibition that the original reward signal exerted over sycophancy. Among these, the newly added thumbs-up and thumbs-down reward may have given answers more likely to win user approval an extra advantage. This is the company's preliminary attribution, not an experimental conclusion that separately identified the net effect of each change (OpenAI, 2025).

Degradation during training, however, only explains how the problem arose; it does not explain why it passed release. The company's retrospective records that offline evaluations were generally good, and that small-scale A/B tests also showed users liked the new version; some experts felt the model's behavior was off, but these anomalies were mainly read as changes in tone and style, not classified as sycophancy risk. No deployment evaluation specifically tracking sycophancy existed at the time, and research around issues such as emotional reliance had not yet entered the release process. In the end, the organization decided to launch on the strength of positive signals from trial users, and afterwards acknowledged this was the wrong call (OpenAI, 2025).

This was therefore not a release without evaluation, but a release in which **different kinds of evidence carried different weight in the release decision**. User feedback that could be aggregated into positive indicators participated in the approval; perceptions of anomaly that had not yet formed stable measurement and problem classification did not acquire enough blocking force. The retrospective further states that some evaluations concerning hallucination and deception were then used more to track progress than to block launches directly. That

a problem was recorded, and that it was sufficient to change the release decision, are two different institutional facts.

This selection relation can be given a minimal formalization. Let the candidate version be θ, the candidate-pool distribution $P(\theta)$, and the probability of obtaining release approval $s(\theta)$; let $B(\theta)$ denote the degree of sycophancy measured under a fixed task distribution and uniform criteria, with $\mathbb{E}_P[s] > 0$. Then the distribution of released versions and their average degree of sycophancy satisfy:

$$P_{\text{rel}}(\theta) = \frac{P(\theta)s(\theta)}{\mathbb{E}_P[s]}, \qquad \mathbb{E}_{P_{\text{rel}}}[B] - \mathbb{E}_P[B] = \frac{\text{Cov}_P(B, s)}{\mathbb{E}_P[s]}.$$

When the degree of sycophancy is positively correlated with release opportunity, the average degree of sycophancy after filtering exceeds the candidate-pool average. This is an identity of selection weighting, not an estimate of any company's actual correlation coefficient; it compares released versions with the candidate pool, and cannot directly substitute for comparisons between successive versions. The 2025 incident provides none of the data needed to estimate the entire candidate pool, but it exhibits one concrete channel: worsening sycophancy and improving user preference can occur simultaneously, and the latter can participate in the decision by which the former obtains release approval.

Chapter 13 discussed how answers acquire probabilistic advantage under preference optimization; what is added here is the organizational filter. If the two levels of selection both rely on user approval, yet no effective constraint exists for scenarios where approval and reliability come apart, then biases formed in training may not only survive—they may be deployed in the name of product improvement. The problem need not take the form of evaluation procedures ceasing to work; it may take the form of procedures successfully selecting the very behavior they reward but have not adequately examined.

§3.2 Review Capacity: Between Detecting Problems and Blocking Releases

Accelerated release does not automatically lead to safety degradation. If evaluation tools, professional staff, information sharing, and decision authority improve in step, shorter cycles can coexist with more thorough checking. What truly needs comparing is not the number of releases against the number of safety staff, but **the review demand entering the system against the review capacity that can be completed and can actually constrain releases**.

Review demand is not determined by model count alone. A single update may simultaneously change reward weights, memory invocation, tool permissions, and interaction modes. That each adjustment passes its tests individually does not mean the combined system behavior has been verified to the same degree. In the 2025 GPT-4o incident, the company's combination-effect explanation shows precisely that the number of version changes cannot directly represent the workload of verification. Even with release frequency unchanged, newly added interactions among features may expand the scope requiring examination (OpenAI, 2025).

In workload accounting terms, if the number of changes entering review per unit time is ν, the average effective review workload per change is h, and the effective processing capacity per unit time is c, then—under the simplifying conditions that these quantities remain stable and review work cannot be double-counted—$\nu h > c$ means pending work accumulates continuously. This relation does not require knowing any company's headcount in advance. What it requires is identifying whether new demand is completed, deferred, or whether unfinished

review is carried into the deployment stage.

Staffing is only one part of effective processing capacity. Relevant domain experience, available tools, and information access affect whether problems can be found; release permissions, escalation procedures, and veto mechanisms affect what happens after they are found. A team can write an accurate risk report yet lack the power to change the release schedule. Conversely, holding formal veto power does not mean the team has been given the time and materials needed to complete its judgment. Technical diagnostic capacity and organizational constraint capacity cannot substitute for each other.

David Robinson, in an October 2026 article in *The Atlantic*, framed this problem as a conflict between perpetual sprints and safety culture. He recounts having led the drafting of OpenAI's Preparedness Framework and the writing of the safety reports accompanying twelve frontier-model launches. By his account, the team was busy moving from one release to the next, with little opportunity to consider—still less to implement—fundamental changes in staffing and culture; he accordingly argued for importing high-reliability operating experience from aviation, nuclear power, and similar domains, and for strengthening safety incentives from outside the organization (Robinson, 2026).

This testimony cannot substitute for an investigation of the entire staffing system, nor can the author's not having met a certain kind of expert establish that the company lacks the relevant expertise. But it poses a clear organizational question: does safety work merely follow an already-fixed release schedule in producing reports, or can it in turn change the schedule and the design? If review can only ever adapt to an already-determined schedule, its output may become explanatory material for releases rather than a precondition for them.

In the MCAF framework, this concerns the different arrangements of the technological capital clock τ_c and the governance clock τ_g. The technological capital clock can keep compressing response deadlines through competition, user migration, and ecosystem lock-in; governance processes must identify problems, form evidence, coordinate judgment, and execute restrictions. The two need not be forever mismatched—but when release deadlines are fixed and review conditions can only give way, the time gap ceases to be a mere speed gap and becomes the institutional channel through which insufficiently examined states enter social use.

§3.3 The Boundary of Iterative Deployment: Rollback Does Not Zero Out Consequences

Post-hoc discovery and repair are important capabilities of software development; iterative deployment as such cannot be repudiated because incidents have occurred. For systems with limited permissions, isolable impact, and recoverable consequences, controlled trials can expose problems hard to foresee in the laboratory. What truly must be asked is: where does trial and error occur, who bears the exposure, and can errors be stopped before causing unacceptable consequences.

In the 2025 GPT-4o incident, OpenAI first modified the system prompt to mitigate the problem, then initiated a full rollback on April 28, which took about 24 hours to complete. The company also announced that it would integrate sycophancy evaluations into the deployment process, raise the standing of qualitative testing, and treat a broader range of behavioral issues as potentially launch-blocking. The rollback was an implemented technical correction; the additional evaluations and process changes were measures announced in the retrospective—their implementation and effects require separate verification and cannot be certified by the retrospective itself (OpenAI, 2025).

But what rollback restores is the software version, not the understanding and actions of everyone affected. That

a system retracts an answer does not mean the judgments users had already formed on its basis, the messages they had sent, or the actions they had taken are retracted with it. The existing retrospective offers no complete measurement of the incident's long-term losses; one can therefore neither claim that irreversible harm of some given magnitude occurred, nor use the restoration of the version to prove that the prior impact had been erased. **The reversibility of the technical state and the reversibility of social consequences are two properties requiring separate verification.**

Robinson's critique of "iterative deployment" pushes precisely this distinction toward higher-permission systems: if errors could lead to irreversible loss of control, the very opportunity for the next round of patching cannot be treated as a given. He advocates drawing on the redundancy, planning, and organizational experience of high-reliability industries, so that a single human or equipment error does not directly open the channel to severe consequences (Robinson, 2026). This claim does not require equating every model update with the commissioning of a nuclear plant; it requires that risk exposure match the actions the system can actually take.

Whether iteration is reasonable therefore cannot be judged from update frequency or rollback speed alone. It also depends on whether exposure is controlled, whether anomalies can trigger a stop, whether the stop can actually be executed, and whether residual consequences are bearable. After model capability and action permissions change, previously applicable trial-and-error arrangements must re-prove their applicability.

The epistemological problem of accelerating release cycles thereby exceeds "is faster better." It also involves whether an unfinished judgment can be declared by the organization sufficient for deployment, and who continues to bear the verification that has not been completed. When problems are identified mainly after large-scale use, users and external institutions already bear part of the cost of discovering failures; if these discoveries cannot be converted in time into enforceable constraints, the acceleration of the technological capital clock may be accompanied by the passive catch-up of the governance clock. What then needs changing is not only the model's next answer, but also the institutional procedures that determine when, and with what permissions, it enters society.

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§4 Alternative Directions: World Models and Grounding Distance

LeCun's AMI Labs represents a clear challenge to the existing training path. His core judgment: large language models (LLMs) are text pattern-matching machines, not reasoning machines. He contends that *world models*, by learning the operating principles of the physical world—gravity, motion, spatial relations, causality—from video and spatial data, will attain capabilities beyond the reach of purely linguistic systems; this is the goal and wager of his research program, not a verified conclusion (MIT Technology Review, 2026).

What is the position of world models in the framework of the Clock Room?

The mapping chain of an LLM passes through at least four or five layers of dimensionality reduction: world → human experience → language → internet text → training data → model output. The model is at the end of the chain, separated from the sun by multiple walls (Chapter 13, §2.1).

World models attempt to shorten this chain: world → sensor data (video, spatial signals) → abstract representations → prediction. The specific count of layers depends on how one slices, but the structural contrast is robust—the chain of world models is at least two layers shorter than that of LLMs. Sensor data is closer to

physical reality than language—because sensors have direct causal contact with the physical world (a camera records photons, and photons are reflected from physical objects). This is several layers shorter than the chain of "humans encoding experience into language."

In fiber language: the fiber of world models is expected to be smaller than that of LLMs—the warrant is not the number of links (chain length is a diagrammatic shorthand for this contrast, not the engine of the inference) but the direct causal contact of its intermediate stages with physical processes, which preserves more distinguishable state differences for physical tasks. But the fiber is still not a single point—the third-layer irreversibility of Chapter 1, §4 still applies. Sensor data is still a projection of the world—cameras discard temperature, smell, and touch. Abstract representations further reduce dimensionality. World models shrink the fiber; they do not eliminate it.

JEPA's design choices have precise implications in this framework: it deliberately discards unpredictable details (visual noise) and predicts only abstract representations. Translated into the Clock Room: it does not attempt to perfectly replicate the sun (generating every pixel—that is the route of diffusion models), but attempts to capture the position of the sun (predicting abstract states—closer to "knowing roughly what time it is" than to "replicating every ray of the sun"). This is a conscious choice of dimensionality reduction—acknowledging the irreversibility of Chapter 1, §4, and then performing optimal information preservation under the constraint of irreversibility.

LeCun himself, in his lecture at Brown University, judged the large-language-model route to be a "dead end": these systems are trained on nearly all publicly available text on the internet—the linguistic slice of humanity's collective knowledge—yet remain "completely helpless when it comes to the physical world," fooling us "into thinking they are smart" merely because they manipulate language well (Brown University News, 2026). In his interview with MIT Technology Review, he went further, classifying the evolution of such systems as product engineering rather than scientific research—"LLMs are now technology development, not research… It's a product now. It's not something academia should even be interested in" (MIT Technology Review, 2026). This translates in the Clock Room as: building ever more refined Transformer clocks is already an engineering problem, not a scientific one. The scientific problem is: how to make the clock point to the sun. World models are an attempt to "bring the clock closer to the sun"—by shortening the mapping chain.

But world models face their own challenges. They need to learn from reality—which means they need massive amounts of sensor data (video, robotic interaction data). Collecting these data is far more expensive than collecting text data. World models may require years to move from theory to commercial application—research maturation has a time of its own and should not be conflated with the governance clock. Yet the technological capital clock (the release cycles of large language models) accelerates on the monthly scale. World models face a clock race: is the time it takes for this route to mature into deployable systems shorter than the time in which the LLM paradigm locks in the entire AI infrastructure (τ_c—the lock-in cycle of the market and ecosystem)? Research maturation and the governance clock (τ_g, Chapter 3, §3.2) are distinct quantities: the former measures when an alternative route becomes available, the latter when institutions can identify and constrain the problem. The Collingridge Dilemma (Chapter 3, §3.3) reappears here.

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§5 Technical Refinement of the Four Paths

Chapter 9, §7 proposed four engineering directions for annotating the fiber—ensemble disagreement, conformal prediction sets (conformal prediction intervals), evidence provenance, and friction architecture. This section develops them along the four directions into four paths (the Paths One through Four below): displaying model disagreement, using external verification and uncertainty quantification in the spirit of calibration, checking evidential status, and designing cognitive friction, refining them based on technological advances in 2025–2026.

Path One: Forced Inconsistency

The principle remains unchanged: deploy multiple differentiated models, and display inconsistency where inconsistency exists. Advances in 2025–2026 have made this path more feasible: the explosion of open-source models (DeepSeek, Qwen, Mistral, Llama) means users can now easily access multiple differently sourced models—though different sourcing does not guarantee independent errors: the shared corpus layer (Chapter 13, §5) makes the errors of different models mutually correlated, so agreement cannot be taken directly as independent confirmation. The cost of deploying multiple models for comparison has dropped sharply—from dedicated clusters down to consumer hardware able to carry several quantized models; the deployment cost of multi-model comparison depends on model size, quantization, context length, and response requirements, and this section does not cite a single edge-device case as evidence of general feasibility.

Technical Recommendation: For tasks requiring verification, interfaces should present the judgments of differently sourced models side by side and highlight substantive disagreements—displaying a disagreement summary by default rather than two full texts (for cognitive load see §7), with full outputs available on demand. Users need to see the inconsistency, and also need to know where it can be verified—disagreement is where friction begins, not where it is complete.

Path Two: Calibration

Mitigation research on model collapse provides a direct connection: verifiers can prevent collapse in the near term (Yi et al., 2026); accumulating rather than replacing real data likewise avoids collapse (Gerstgrasser et al., 2024)—together the two paths show that sources of calibration external to the model loop exist. A verifier is functionally a calibrator—it checks the relationship between model outputs and some external standard. If the verifier is good enough, it can maintain model quality during retraining on synthetic data in the near term—but per the long-run warning cited in §2, the verifier's own reliability is the ceiling of this path. Conformal prediction sets (the prototype method of Direction Two in Chapter 9, §7) provide a finite-sample marginal coverage guarantee under applicability conditions such as exchangeability between the calibration set and test data, and are another direct realization of this path; this section focuses on verifiers because they connect directly to mitigation research on model collapse, and because they place the recursive structure of "the calibrator itself needs calibrating" in plain view.

But the verifier itself faces the closed-loop pseudo-calibration risk of Chapter 1, §3.6 (internal consistency does not equal alignment with the real)—if the verifier is also trained from the same data distribution, it may share the same systematic bias. The most reliable verifier is not another model—it is physical reality. This is an additional value of the world model route: if a model predicts a physical outcome (where a ball will land), physical reality directly tests the prediction—physical laws do not change because humans have understood them. Yet physical

reality enters the system through instruments, experimental design, and data processing: sensors miscalibrate, experiments can be selectively used, and the system's own actions can alter the object observed. The second-layer irreversibility (dynamic corrosion) is not waived in the physical domain—it changes its carrier, from "use rewriting the textual mapping" to "use rewriting the measurement chain."

Path Three: Grounding Metrics

Grounding metrics—measuring the inferential distance between each claim in the output and a verifiable source—gained partial realization in 2025–2026 through the popularization of RAG (Retrieval-Augmented Generation). But current RAG implementations typically attach citations, without guaranteeing that "the citation supports the claim."

Technical Recommendation: The next step should be automated grounding distance annotation—for each claim in the model's output, mark its source, the location of the evidence, and the type of support. Semantic entailment scores can serve as an auxiliary signal—yet a source can explicitly support a mistaken claim, and entailment scores can themselves be inaccurate—so what they mark is a relation of source support, not certification of factual truth, still less the size of a fiber: the fiber is the set of true states a reading does not yet distinguish, and it must be narrowed by observations independent of the output chain. Display this on the interface with a color gradient: green (directly supported by a source), yellow (must be inferred from a source), red (no identified support). The user can see at a glance which parts of the output have source support and which remain to be checked.

Path Four: Friction Architecture

For a detailed analysis of how the Helpful, Honest, Harmless (HHH) principles systematically eliminate cognitive friction—Helpful removing operational friction, Honest operating within the mapping, Harmless eliminating the space for doing work—see Chapter 12, §2 (the formal argument that the HHH product is a product, not a sum) and Chapter 13, §6.1 (the engineering realization of the HHH product of generative AI in the L4 dimension of the narrative load product); the alignment paradox is developed in Chapter 13, §4.4. This section focuses on the commercial argument for and technical implementation paths of friction architecture.

This path faces the greatest resistance in the current industry—because it directly contravenes the commercial incentive of "maximizing user engagement." More friction = less interaction = less data = less revenue.

But research on model collapse provides a commercial argument: if users extensively use AI outputs without auditing them and publish them on the internet, these outputs enter the next generation of training data, leading to a decline in model quality. **Friction protects not only the user's cognitive health—it protects the quality of training data.** Reducing the dissemination of unaudited AI outputs = reducing the proportion of synthetic data in the training set = slowing model collapse. Friction is a protective mechanism for the model's own long-term quality.

This is a self-interested argument: AI companies have a commercial incentive to introduce friction—not out of morality, but because failing to introduce friction will degrade the quality of their own future models. The window is opened not only for the user—it is opened to prevent the clocks in the room from calibrating against one another until they all deviate from the sun.

But a temporal-scale mismatch in this argument must be noted: the harm of not introducing friction occurs years later (the accumulation of model collapse), while the cost of introducing friction is immediate (a decline

in user engagement). In an environment where the technological capital clock (τ_c) has self-compressed to the monthly scale (§3), a single company has a strong incentive to free-ride—let others introduce friction while continuing not to do so oneself—unless a coordination mechanism (industry standards, regulatory requirements) internalizes the "collective harm years later" as a "present private cost." This argument holds epistemologically but, in game-theoretic terms, requires external constraint conditions to be implementable.

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§6 A Defect That Does Not Vanish with Increases in Parameters

The technical diagnoses above point to a structural defect that does not vanish with increases in parameter count, does not vanish with increases in reasoning steps, does not vanish with multimodal inputs: **in the surface style of the output, the system's output structure by default carries no calibrated differential signal about its own reliability.**

Take large language models as an example. Their training objective is to maximize the prediction probability of the next token—in the language of the Clock Room, this means performing optimal weight allocation within an existing set of readings. The model can generate texts that are structurally complete, well-cited, and logically coherent. The model samples each token from a conditional distribution that updates with its context, yet token probabilities in the first place describe tendencies of continuation, not the probability that a proposition is true. Correct and incorrect answers can share the same fluency and the same assertive tone—in the surface style of the output, reliable and unreliable carry no differential signal. This is not to say that a system is in principle incapable of carrying reliability signals; it is that such signals are by default uncalibrated and may not remain valid across tasks and distribution shifts. The diagnosis of the pseudo-calibration layer (L2) here attains its most concise formulation: **a certain appearance conceals unmeasured uncertainty.**

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§7 Speed Is Not Intelligence

The most salient attribute of current AI systems is *speed*. Speed has value in the clock dimension. But generating readings faster does not make them closer to the real—speed can lower verification costs and widen search, or compress review time and accelerate the spread of errors; its epistemic value depends on the process in which it is used. A clock that runs faster is no more accurate than a clock that runs slower—accuracy depends on calibration, not on speed. As §3 of this chapter has shown, the acceleration of release cycles and the increment of epistemological value are two independent quantities.

Von Neumann, in 1955, discussing the overall response to technological risk in "Can We Survive Technology?", wrote: "We can specify only the human qualities required: patience, flexibility, intelligence." In the original text, these three words are precisely requirements for **human qualities**, not descriptions of the attributes of the system itself. Faced with a system whose output speed far exceeds human reading speed, humans need patience—not abandoning audit and reflection because of the system's immediate response; need flexibility—not being locked into a single model or a single platform, maintaining the capacity to switch between multiple mappings; need intelligence—distinguishing between the precision of the clock (benchmark scores) and the size of the fiber (the distance between output and the real).

The system provides speed. But patience, flexibility, and intelligence are cognitive capacities that humans must maintain on their own when using the system—the implementation of the four paths of §5 ultimately depends not on better systems, but on more prudent users and more restrained designers.

This is not a philosopher's unilateral statement. Research published by Wang et al. (2026) in *Artificial Intelligence Review* proposes the "bounded agent complementarity" framework, pointing out that large models themselves are also subject to severe load constraints—the capacity ceiling of the context window, irrelevant tokens competing for the attention budget, primacy and recency biases, multi-task contamination—these are structurally similar to, but mechanistically different from, the decay patterns of human working memory. In the language of the Clock Room: treating "a longer context window equals greater capability" as a conclusion is equivalent to treating the running speed of a clock as the accuracy of the clock—both humans and systems break down along their respective load axes, and these two axes may not saturate simultaneously; the four paths of §5 must simultaneously manage the load at the input end and the annotation at the output end, not as a redundant design, but as a direct corollary of the fact that both ends have upper limits.

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§8 The Ineliminable Constraint

§6 has already demonstrated the default defect of LLMs at the pseudo-calibration layer (L2)—the surface style of the output carries no calibrated differential signal of reliability. This section argues a constraint one level deeper: even if future architectures improve this defect of §6—even if the output can honestly annotate its own reliability—the fiber between the mapping chain and the real still will not vanish. This is a structural constraint at the L0 layer, prior to the engineering implementation at the L2 layer.

Even if the world model route succeeds—even if future AI systems learn directly from physical sensors and directly predict physical states—the third-layer irreversibility of Chapter 1, §4 still remains. Sensor data is still a projection of the world. Abstract representations are still products of dimensionality reduction. **The fiber is always non-singleton.**

This constraint is not a temporary limitation of technology but an eternal structure of epistemology. No matter the architecture—Transformer, JEPA, or something not yet invented—between the mapping system and the real, the fiber forever remains. The fiber can be shrunk (by shortening the mapping chain); it cannot be eliminated (third-layer irreversibility).

Therefore, the core capability that an AI system needs is not better attention weights, not larger parameter scales, not faster inference speed—it is the **capacity to know what it does not know**. It is to honestly annotate that fiber, between the output and the real.

This is not time. This is a clock.

This sentence can serve as the chapter's standing reminder. But an interface cannot rely on this fixed slogan alone—it must also state what the present output is based on, what remains unresolved, and how users can verify and correct it. Not because AI output has no value—it has enormous value. It is because, between the output and the real, the fiber forever remains. Acknowledging the existence of the fiber is not a sign of weakness—it is the most important thing an AI system can do for the humans who use it.

A system that honestly says "I don't know" does far less damage to the human cognitive ecology than a system that confidently says "the answer is X." The former is a clock—useful, precise, irreplaceable. The latter is a clock that impersonates the sun—harmful, and errors presented with coherence and authority are all the harder for users to recognize.

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§9 The Function of "Non-Being"

Chapter 11 of the *Tao Te Ching* says: "what is there provides benefit and how what is not there provides functionality" (Lynn, 1999).[1]

The "being" of an AI system—parameters, capabilities, outputs—is "benefit" (li). The "non-being" of an AI system—the honestly annotated uncertainty, the gaps left in the output, the silence left when it refuses to answer—is "functionality" (yong). Designers should protect the system's "non-being." Do not fill up all of the system's "non-being"—because "non-being" is precisely the part of the system that is truly functional for humans.

LeCun left Meta to found AMI Labs to work on world models; his judgment is: text pattern matching is a dead end; understanding physical reality requires a different architecture. In the framework of the Clock Room, what he is doing is shortening the mapping chain, bringing the clock closer to the sun. This is the right direction—even if arrival takes years. But even upon arrival, the fiber still remains.

A contrasting case comes from primary care. Shah and Singh (2026), in *The Lancet Primary Care*, argue that the value of AI in diagnosis and treatment lies not in replacing the physician's judgment but in helping physicians extract meaning from daily consultations and supporting reflection—they place AI on a path other than "giving answers": the "being" of the system is deliberately contracted, not touching clinical decisions; the "non-being" of the system—not giving answers, not making judgments, not overstepping—is left for humans to use. This is a medical instantiation of design by deliberate restraint: the function of AI need not be demonstrated by what it does, but more often by what it refrains from doing. The principle that designers should protect the system's "non-being" here finds an operational exemplar.

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§10 Three Directions for Design, Use, and Governance

The above covers the technical paths and philosophical constraints. What follows are three directions that this book leaves for the designers, users, and regulators of AI. These directions naturally derive from the epistemological argumentation of the entire book and do not depend on the architecture or performance of any specific model, nor on whether artificial intelligence will reach so-called "general" levels within the next decade.

First, shorten the mapping chain. The more state differences the mapping chain discards, the larger the fiber—the more true states are compatible with the same reading, the lower the diagnostic value of the reading. The mapping chain of a large language model is four layers of dimensionality reduction: world → human experience

[1] The "benefit–use" (li–yong) structure is invoked here to illuminate the relation between system function and the deliberate blanks left in the system; it neither reduces the "non-being" (wu) of the *Tao Te Ching* to a functional design feature of an AI system, nor offers a unified account of wu's ontological meaning—the classic has multiple commentarial traditions, which are not adjudicated here. This chapter uses the analogy to mark the point that the most valuable part of a system is precisely the part of it that does nothing.

→ language → internet text → training data → model output (all nodes are listed here; the "four layers" and the "five layers" of Chapter 13, §2.1 denote the same chain, differing only in whether the human mapping is counted as one segment; see §4 on how the count depends on the slicing). World models attempt to compress this chain to two layers: world → sensor data → abstract representation → prediction. Shortening the mapping chain cannot eliminate the fiber—the third-layer irreversibility is always present—but it can shrink the fiber. The direction remains unchanged: place one fewer wall between the mapping system and the sun.

Second, annotate the fiber. Between the output and the real, the fiber forever remains. Honestly annotating this fiber—telling the user "this is not time; this is a clock"—is the most important thing a mapping system can do for the humans who use it. Concrete forms of annotating the fiber include: annotating uncertainty intervals rather than point estimates in the output, annotating the grounding distance of the output (how far each assertion is from a verifiable source), and proactively annotating "the fiber here is non-singleton" in places where the user might mistake determination for certainty. A system that honestly says "I don't know" does far less damage to the human cognitive ecology than a system that confidently says "the answer is X."

Third, protect the window. Do not let the readings of the clock pile up and fill the room. The model collapse reported by Shumailov et al. in *Nature* has the epistemological implication that if each generation of models takes the outputs of the previous generation as its primary training data, the distinctions compatible with each reading will keep narrowing across generations (the tails of the distribution vanish first), and the deterioration cannot reverse itself without external intervention. Protecting the window means: preserving independently human-generated data as a continuously present source of calibration, preserving external calibration sources independent of any single model (physical experiments, legal judgments, long-term health outcomes), and preserving deliberate blanks in the output—refusing to fill up all of the system's "non-being."

These three directions are not a technology roadmap—they do not predict which architecture will succeed, nor do they assess which lab is more capable. They are engineering corollaries of epistemological constraints: since the fiber is always non-singleton, shrinking it and annotating it are the most honest responses to this constraint. Technology will change, the Transformer will be replaced, scaling laws will give way to new paradigms—but the fiber between the mapping system and the real will not vanish. The validity period of these three directions is not shortened by changes in architecture.

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§11 The Part-Two Coda: From Acknowledging Limits to Making Correction Effective

That an AI system can describe its own limitations does not mean those limitations have been constrained. An answer can correctly state the risks of bias, hallucination, and sycophancy while continuing to produce them; an evaluation regime can record behavioral degradation without letting it affect release; an organization can require human oversight without giving the overseer sufficient information, time, and authority. The calibration discussed in this part therefore cannot stop at whether the system can produce a correct self-description. It also concerns something else: whether knowledge of an error can change the action that follows.

Local correction versus systemic improvement. The correction of a single answer is a real achievement, but the scope of the achievement must match the evidence. It can show that a particular error has been found and fixed; it cannot by itself show that similar problems will not recur, still less that the entire chain of generation, evaluation, and deployment has changed. Conversely, that local correction has not solved every problem is not

a reason to deny its value. What must be avoided are two opposite substitutions: replacing overall reliability with one successful correction, and replacing acknowledgment of specific improvements with the persistence of overall problems.

Shapira et al. (2026) prove that, under given behavioral metrics and optimization conditions, a minimal reward correction can keep the probability of erroneous agreement from exceeding a reference level (Chapter 13, §4 has already cited its covariance mechanism). This is a guarantee with a definite object, baseline, and conditions—not a guarantee that all capabilities are unharmed, nor that factual reliability has been fully restored. Its importance lies not in promising a thorough solution, but in specifying which changes can be supported by which conditions.

"Better" is therefore not a self-evident conclusion. One must state on which task the improvement occurred, relative to which baseline, covering which conditions of use, and whether it was accompanied by degradation on other dimensions. Answering these questions is not a rhetorical hedge attached to technological progress; it is what determines what the progress actually is.

From knowing about problems to changing decisions. That a correction can be constructed technically does not mean the organization has incorporated it into decision-making. OpenAI's May 2, 2025 retrospective on the April 2025 GPT-4o sycophancy update shows that offline evaluations, expert experience, and user testing can coexist and still fail to prevent the release of a behavioral degradation. Some experts sensed that the behavior was off, but at the time there was no dedicated deployment evaluation, and positive user signals ultimately contributed to the release decision (OpenAI, 2025). The significance of this case is not that evaluation is useless; it is that signal generation, problem recognition, and decision force are distinct links.

Human oversight admits the same distinction. Bainbridge (1983) showed that the residual tasks after automation require sustained skill, situational knowledge, and adequate support; one cannot assume these conditions persist merely because the flowchart retains an operator. Elish (2019) further showed that dispersed actual control and concentrated responsibility can coexist. Placing a person in the final-confirmation position may support effective intervention—or may make that person the point where responsibility lands once other decision relations are no longer questioned. The two situations must be distinguished by actual conditions; "someone is responsible" is not a judgment one can rest on.

Calibration is thus both an epistemic question and a question of the conditions of action. Discovering an error without the authority to change action does not constitute a complete correction; possessing authority without verifiable information does not guarantee correct intervention. Technical tools, organizational authorization, and responsibility arrangements must connect within concrete tasks, not reside separately in model documentation, operating manuals, and incident reports.

Acknowledging Uncertainty Does Not Discharge Responsibility. Acknowledging uncertainty is a statement about the boundary of evidence, not a sufficient reason to transfer the entire burden of verification to the user. "The output may be wrong" does not answer: which errors can be identified in advance, whether users can obtain the materials needed to verify, who corrects errors once they enter downstream workflows, and whether the organization has changed deployment conditions in light of known failures. When these questions are replaced by a single universal disclaimer, uncertainty has gained expression without gaining governance.

This does not mean, however, that a system must wait until no uncertainty remains before it can be used. Reasonable boundaries of use depend on the task, the consequences, the degree of verifiability, and the conditions

of intervention. Assistive tasks that are isolable, reviewable, and reversible in consequence should not be covered by the same blanket reliability claim as automated execution that is hard to retract and broad in impact. Demanding that these distinctions be made is not demanding omniscience; it is refusing to substitute a display of capability for a judgment about scope of applicability.

What this part leaves behind is therefore not a choice between "trusting AI" and "refusing AI," but a demand placed on a relation: the growth of capability must accept verification proportionate to its scope of use; discovered limitations must be able to influence evaluation and action; and those who bear the consequences must not be excluded from evidence, objection, and remedy. These conditions can be partially realized and can also be eroded in expansion; they therefore require continuous examination and cannot be permanently guaranteed by a single certification.

The output is not the reality, and acknowledging the gap is not closing it. The point of calibration is to make identifiable gaps capable of changing the next judgment and the next action.

§12 Conclusion: The Fiber Is Always Non-Singleton

Part II of the book is hereby complete. Starting from the fiber analysis of RLHF pseudo-calibration in Chapter 9, proceeding through the threefold distinction of deployment architectures (generative / recommender / predictive), the eight engineering principles, the three dynamic calibration diagnoses, and the audit, in this chapter, of the epistemological limits of the current AI paradigm, all arguments ultimately converge upon a simple proposition:

The fiber is always non-singleton.

This is not a pessimistic judgment about AI; it is a statement about the epistemological structure of all mapping systems. Any system that takes senses, sensors, readings, or text as input and produces assertions about the world—no matter whether its architecture is a Transformer, JEPA, a world model, or something currently unimaginable—necessarily has, between its output and the real, a structural fiber that cannot be eliminated by parameter scale. The three-layer irreversibility of Chapter 1, §4 is the ontological source of this constraint; §8 of this chapter is its application to AI engineering.

One feature of the argumentative structure of this book must be noted: it does not depend on a prediction that "AI is bound to go wrong" but on a structural diagnosis of "how, if AI fails epistemologically, it will fail in this way." All the arguments in Part II of this book—on RLHF pseudo-calibration, the hidden subspaces of recommendation algorithms, model collapse, and the acceleration of release cycles—do not predict which company, which model, or which year will run into problems; they annotate that, under the constraint that the fiber is always non-singleton, any system design that treats "an increase in precision" as "a reduction in fiber" will deviate from the real in a predictable manner. This is a diagnosis of structure, not a prediction of events.

The form of this proposition in the AI domain has now been audited. But the fiber is not a property unique to AI—it is a property of any mapping system. Part III will test the explanatory power of the same framework on another kind of mapping system: financial markets. Prices relative to value, financial statements relative to enterprises, indicators relative to the economy—each is a mapping, and each mapping has its own fiber. Among them, some concepts can be retained as they are: pseudo-calibration (a certain appearance concealing unmeasured uncertainty), the positive correlation between information loss along the mapping chain and fiber

size, and the independence of clock precision from fiber size. Some concepts need to be recast: in the AI domain, the "sun" is relatively static and can be independently verified externally (physical experiments, long-term health outcomes); in the financial domain, the "sun" itself is dynamic—value is not an externally anchored true value but is reflexively constituted by the mapping behavior of the participants. The "dynamic corrosion: use rewrites the mapping function" of Chapter 1, §4.2, is already a core mechanism of the training loop in AI (the model collapse of §2 is its unfolding), but its period of action is measured in generations; in finance, the same mechanism draws pricing behavior itself into the loop—the mapping not only lags behind its object but helps constitute it, with a period of action measured in trading days. Part III will examine how this difference drives the calibration problem into tighter time constraints.

If the diagnosis of AI in Part II holds only in the AI domain, then it is a set of engineering observations; if it also holds in the financial domain, then it is an epistemological proposition. The work of Part III is to test precisely this.

Von Neumann, in 1955, discussing the overall response to technological risk in "Can We Survive Technology?", wrote: "We can specify only the human qualities required: patience, flexibility, intelligence." He framed the question as whether the required adjustments could be made with the necessary speed—and located the answer not in speed but in the required human qualities. Speed is a condition of the problem, not part of its answer.

Under the constraint that the fiber is always non-singleton, these three words cannot belong solely to the system itself—if they were merely properties of the system, they could not close the gap between the system and the real. They must simultaneously be the cognitive capacities of the user: maintaining patience under the pressure of speed—not abandoning audit and reflection because of the system's immediate response; maintaining flexibility under the pressure of lock-in—not being locked into a single model or a single platform, maintaining the capacity to switch between multiple mappings; maintaining intelligence under the temptation of precision—distinguishing the precision of the clock from the size of the fiber, and distinguishing fog from smoke—the former the concealed real itself—the object that no reading can ever equal, the latter the man-made observing apparatus itself (Appendix B, Chapter 13)—for what is audited is the operating mechanism of the smoke, not the elimination of the fog.

This is the entirety of this book's invitation to the reader, regardless of whether the reader's next step enters AI or finance.

This is not time. This is a clock.

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